Shipeng Liu

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Research interests

Embodied "Tactile Sensing", Robot Planning and Navigation, Embodied AI Agents, Proprioceptive Feedback, Robot Co-scientists

- My research focuses on building robots with **embodied tactile sensing capabilities** enhanced by proprioceptive feedback, advancing intelligent robot planning and navigation systems, and developing embodied AI agents and robot co-scientists for autonomous exploration. I develop novel sensing frameworks that enable robots to achieve tactile perception and adapt to complex environments through proprioceptive feedback, create intelligent planning algorithms for **robust navigation** across diverse terrains, and design **embodied AI systems** that collaborate with humans in scientific discovery and field exploration tasks.

Education

University of Southern California — Los Angeles, CA
PhD in Electrical Engineering — 09/2021 – 03/2026 (Expected)
Advisor: Dr. Feifei Qian

University of Southern California — Los Angeles, CA
Master in Electrical Engineering — 09/2021 – 09/2025
Advisor: Dr. Feifei Qian

Tongji University — Shanghai, China
Bachelor of Engineering in Electrical Engineering (with Honors) & Minor in Applied Math — Awarded in 2020

Honors and scholarships

MHI Scholar Award Finalist (University of Southern California)	2024
USC Top-off Fellowship (University of Southern California)	2024
CURVE PhD Mentor Award (USC)	2025, 2024, 2023
Best Paper Finalist (HRI 2024)	2024
WiSE Travel Award (University of Southern California)	2024, 2022
Provost's Fellowship Graduate School Fellowship (USC)	2021
Outstanding Graduate Student (Tongji University)	2020

Publications

In submission

[1] Sean A. Buchmeier, Freya Whittaker, Shipeng Liu, Ian C. Rankin, and Cristina G. Wilson, Supervisory Control for Human-Robot Teaming: Insights from a Field Study in Scientific Data Collection.

[2] Ian C. Rankin, Shipeng Liu, Freya Whittaker, Sean A. Buchmeier, Liam Bouffard, and Cristina G. Wilson, Robot-Assisted Exploration Decisions during a Planetary Analog Field Science Campaign.

[3] Shipeng Liu†, Diego Caporale†, Ethan Fulcher, Wilson Hu, Natalie Cavallo, Yifeng Zhang, Xingjue Liao, Cynthia Sung, and Feifei Qian, Scout-Rover cooperation: online terrain strength mapping and traversal risk estimation.

[4] Shipeng Liu, FNU Shrutika, Boshen Zhang, Zhehui Huang, Gaurav Sukhatme and Feifei Qian, Effect of Adaptive Communication Support on LLM-powered Human-AI Collaboration. [Arxiv]

[5] Shipeng Liu†, Boshen Zhang†, Zhehui Huang, Feifei Qian, Gaurav Sukhatme, Towards Real-time Adaptation of Embodied Agent in Human-Robot Collaboration. [Arxiv]

[6] Shipeng Liu, Meghana Sagare, Shubham Patil, Feifei Qian, Bio-inspired tail oscillation enables fast crawling on soft sand. [Arxiv]

[7] Matthew Jiang†, Shipeng Liu†, Feifei Qian, Safe Active Navigation and Exploration for Planetary Environments Using Proprioceptive Measurements, [Arxiv]

[8] Yimin Tang†, Shipeng Liu†, Sven Koenig, Discrete Multi-Agent Path Planning via Discrete Categorical Diffusion with Constraint Projections

Peer-Reviewed Journal Articles

[9] Zhen Zhang, Jintao Lai, Fangkai Wang, Xiaoguang Yang, Shipeng Liu, Mingyu Zhang, A Trajectory-Based Control Strategy with Vehicle Cooperation and Absolute Transit Priority at an Isolated Intersection, Journal of Advanced Transportation 2024. [Link]

[10] Shipeng Liu, Cristina G. Wilson, Bhaskar Krishnamachar, Feifei Qian, Understanding human dynamic sampling objectives to enable robot-assisted scientific decision-making, ACM Transactions on Human-Robot Interaction (THRI), 2023. [Link] (JCR, Q1, IF: 5.5)

[11] Shipeng Liu, Boyuan Huang, and Feifei Qian, Adaptation of flipper-mud interactions enable effective terrestrial locomotion on muddy substrates, IEEE Robotics and Automation Letters (RAL), 2023. [Link] (JCR, Q1, IF: 5.3)

Peer-Reviewed Conference Proceedings

[12] Rachel Holladay, Shivangi Misra, Mason Mitchell, Akshay Ram Panyam, Erica Pauline Santos, J. Diego Caporale, Shipeng Liu, Yifeng Zhang, John Ruck, Douglas Jerolmack, Feifei Qian, Mark Yim, and Cynthia Sung. Characterizing Robot-Ground Interactions for Autonomous Lunar Construction, 19th International Symposium on Experimental Robotics (ISER'25).

[13] Shipeng Liu, Siyuan Meng, Jiaze Tang, and Feifei Qian, Proprioceptive Sensing Enabled Granular Media Estimation for Locomotion Adaptation, Robotic Science & Systems (RSS). [Acceptance rate: 25%]

[14] Shipeng Liu, Cristina Wilson, Zac Lee, Feifei Qian, Modelling experts' sampling strategy to balance multiple objectives during scientific explorations, In 2024 ACM International Conference on Human-Robot Interaction (HRI' 24) [Link] [Best Paper Finalist 5%]

[15] Shipeng Liu, Boyuan Huang, and Feifei Qian, Adaptation of flipper-mud interactions enables effective terrestrial locomotion on muddy substrates, IEEE International Conference on Robotics and Automation (ICRA), 2024. [Link]

[16] Xinyue Chen, Shuo Li, Shipeng Liu, Robin Fowle, and Xu Wang, Meetscript: Designing transcript-based interactions to support active participation in group video meetings, in ACM Conference on Computer-Supported Cooperative Work & Social Computing (CSCW 2023). [Link]

Workshops and Abstracts Presentation

[17] Matthew Jiang†, Shipeng Liu†, Feifei Qian, Safe Active Navigation and Exploration for Planetary Environments Using Proprioceptive Measurements, MOO workshop in Robotic Science & Systems (RSS), June 2025 - Poster.

[18] Shipeng Liu†, Diego Caporale†, Ethan Fulcher, Wilson Hu, Natalie Cavallo, Yifeng Zhang, Xingjue Liao, Cynthia Sung, and Feifei Qian, Scout-Rover cooperation: online terrain strength mapping and traversal risk estimation, 56th Lunar and Planetary Science Conference (LPSC), Mar 2025 - Poster.

[19] Shipeng Liu, Jiaze Tang, Siyuan Meng, and Feifei Qian, Proprioceptive Terrain Characterization and Adaptive Locomotion on Muddy Surfaces, March Meeting of the American Physical Society, Los Angeles, CA, 2025 - Oral.

[20] Deanna Flynn, Shipeng Liu, Feifei Qian, and Cristina G. Wilson, Explicable and Explainable Robot Sampling Suggestions: A Field Demonstration with an Expert Geoscientist, In workshop of the 2024 ACM International Conference on Human-Robot Interaction (HRI' 24) - Poster.

[21] Shipeng Liu, John C. Bush, Shravan Pradeep, Douglas J. Jerolmack, Siyuan Meng, Jiaze Tang, John Ruck, and Feifei Qian, Proprioceptive sensing to aid with locomotion adaptation in mud, March Meeting of the American Physical Society, Minneapolis, MN, 2024 - Oral.

[22] John C. Bush, Yifeng Zhang, John D. Caporale, Robert C. Ewing, Kristin R. Fisher, Eric Fulcher, Douglas J. Jerolmack, Nathan A. Jones, Daniel E. Koditschek, Shipeng Liu, Brett McKeeby, Medhavi Nachan, Felipe Rivera-Hernandez, John Ruck, Thomas Shipley, Sheryl Thompson, Cristina Wilson, and Feifei Qian, Relating Geotechnical Properties of Crusty Regolith to Morphology and Mineralogy Using a Robotic Leg Rheometer, Lunar and Planetary Science Conference (LPSC), Woodland, TX, 2024 - Oral.

[23] Yifeng Zhang, Eric Fulcher, John D. Caporale, Shipeng Liu, John Bush, John Ruck, Daniel E. Koditschek, Douglas J. Jerolmack, and Feifei Qian, Making every step an experiment: proprioceptive sensing during locomotion for enhanced mobility and data collection in earth and planetary explorations, March Meeting of the American Physical Society, Minneapolis, MN, 2024 - Oral.

[24] John Bush, Yifeng Zhang, Jake Futterman, John Ruck, Shipeng Liu, Eric Fulcher, Douglas J. Jerolmack, Kristin R. Fisher, Robert C. Ewing, and Feifei Qian, Characterizing Mechanical Properties of Natural Deformable Substrates with a Direct-Drive Robotic Leg, March Meeting of the American Physical Society, Minneapolis, MN, 2024 - Oral.

[25] Ethan Fulcher, John Bush, John D. Caporale, Shipeng Liu, Yifeng Zhang, and Feifei Qian, Robotic Legs as Novel Regolith Sensors for Earth and Planetary Explorations, Southern California Robotics Symposium, Los Angeles, CA, 2023 - Oral.

[26] John Bush, Yifeng Zhang, Shipeng Liu, Ethan Fulcher, Cecil Franklin, Douglas J. Jerolmack, Ryan C. Ewing, Kenton R. Fisher and Feifei Qian, Robotic legs as novel planetary instrumentation to explore the mechanical properties of regolith, 54th Lunar and Planetary Science Conference, 2023 - Poster [Dwornik Best Poster Presentation Award runner-up].

[27] Boyuan Huang, Shipeng Liu, Jake Futterman, and Feifei Qian, Flipper-based Locomotion on Muddy Substrates, Bulletin of the American Physical Society, 2023 - Oral.

[28] Cristina Wilson, Shipeng Liu, John Ruck, Douglas J. Jerolmack, Thomas F. Shipley, Mingzhuo Li, Zeyu Wang, and Feifei Qian, Human-robot Teaming for Real-time Data Foraging Decisions, AGU Fall Meeting Abstracts, vol. 2022, pp. IN45D-0396, 2022 - Poster.

[29] Shipeng Liu, Cristina Wilson, Bhaskar Krishnamachari, and Feifei Qian, Human Geoscientist Objective Functions for Robot-Aided Field Data Collection Decisions, ICRA Workshop, 2022 - Poster.

[30] Shipeng Liu, Cristina Wilson, Thomas Shipley, Daniel Koditschek, and Feifei Qian, Understanding Human Decision Making in Robot-Assisted Field Data Collection, AGU Fall Meeting Abstracts, vol. 2021, pp. EP55A-1040, 2021 - Poster.

Research experience

Robot Locomotion and Navigation Dynamic Lab

Research Assistant 09/2021 – Present
Topics: Robotics co-scientists for Planetary Exploration

Lifelong Learning Lab
Research Assistant 06/2020 – 09/2020
Topics: Building AI tools for Online Meetings

Industry experience

SenseTime, Autonomous Driving Division Shenzhen, China

Autonomous Driving Engineer 09/2021 – 09/2021
Topics: decision-making, lane-change algorithm of autonomous driving

Unity-Drive, Autonomous Driving Division Shenzhen, China

Autonomous Driving Intern 05/2019 – 09/2019
Topics: simulation development based on Carla

Teaching experience

Teaching Assistant, ECE, USC Fall 2024

EE510: Linear Algebra
Topics: Linear algebra fundamentals, matrix operations, vector spaces, eigenvalues and eigenvectors.

Teaching Assistant, ECE, USC Spring 2024
EE559: Machine Learning I: Supervised Methods
Topics: Supervised learning algorithms, classification, regression, model evaluation.

Teaching Assistant, ECE, USC Fall 2023
EE599: Robotic Mobility
Topics: Robot locomotion, control systems, terrain adaptation, bio-inspired robotics.

Teaching Assistant, ECE, USC Spring 2023
EE559: Machine Learning I: Supervised Methods
Topics: Supervised learning algorithms, classification, regression, model evaluation.

Teaching Assistant, ECE, USC Fall 2022
EE599: Robotic Mobility
Topics: Robot locomotion, control systems, terrain adaptation, bio-inspired robotics.

Teaching Assistant, College of Design and Innovation, Tongji Fall 2018
Open Source Hardware and Programming course
Topics: Open source hardware, programming fundamentals, project development.

Mentoring experience

⋆ co-authors of publications and abstracts; & underrepresented groups

Graduate Students
⋆&**Meghana Sagare** (Next Position: PhD student at UCI) 04/2024 – 10/2024
&**Harshita Arya** (Next Position: Machine Learning Engineer at Zefr) 09/2023 – 06/2024
⋆&**Shrutika Shrutika** (Next Position: Software Engineer at Meta) 09/2023 – 09/2024
- **Bredan Hy** (Next Position: Software Engineer at Sentry) 09/2023 – 01/2024
⋆**Jiaze Tang** (Next Position: PhD student at UCR) 01/2023 – 03/2025

⋆**Siyuan Meng** (Next Position: PhD student at Rutgers) 01/2023 – 03/2025
⋆**Zeyu Wang** (Next Position: Software Engineer at Nucor) 06/2022 – 06/2023
⋆**Boyuan Huang** (Next Position: PhD student at UCR) 12/2021 – 01/2024
⋆**Sen Gao** (Next Position: Ph.D. student at University of Colorado Boulder) 12/2021 – 05/2023
- **Mingzhuo Li** (Next Position: Software Engineer at Empyrean Technology) 09/2021 – 01/2022
⋆**John Bush** (Next Position: Embedded Software Engineer at Impulse Space) 09/2021 – 06/2024

Undergraduate Students

&**Helena Teixeira-Dasilva**	09/2025 – Present
&**Tarunika Ponnusamy**	06/2025 – 08/2025
- **Michael Leong**	01/2025 – Present
&**Kaylee Hou**	10/2024 – Present
&**Cynth Liu**	10/2024 – Present
- **Seojoon Kwon**	10/2024 – Present
- **Abhigya Goel**	10/2024 – 01/2025
&**Priscilla Ortiz-Zamora**	07/2024 – 08/2024
- **Lucas Dudley**	08/2023 – 05/2024
- **Shantanu Jhaveri**	01/2023 – 05/2023
⋆**Matthew Jiang**	01/2023 – Present
⋆**Jake Futterman**	12/2021 – 06/2022
&**Jessica Yuan**	09/2021 – 12/2021

High School Students

Derek Kong (Next Position: Undergraduate student at University of Pennsylvania) 05/2023 – 09/2023

Skills

Programming

- Proficient in: Python, C/C++, Matlab, JavaScript, React.js, C#/Unity.

Software

- Carla simulator, Webots, Gazebo, ROS2, Solidworks, Linux, Embedded system programming, Principal Component Analysis, SVM, K-Means, Version control (git), Reinforcement Learning, Inverse Reinforcement Learning, Diffusion model for path/action planning, CUDA, PyTorch, Transformer

Hardware

- Odrive, Motor control board, Raspberry Pi, Arduino

Other Skills

- LLM prompt engineering, LLM fine-tuning (LoRA), Project organizing, Mathematical Modeling, RAG (Retrieval-Augmented Generation), LangGraph

Languages

English (Fluent), Chinese/Mandarin (Native), Chinese/Shanghainese (Native)

Selected Coursework

Advanced Robotics: Computational Human-Robot Interaction, Robot Learning, Deep Learning for Manipulation
Learning: Machine Learning, Advanced Computer Vision, Probabilistic and Generative Models

Developed Software and Robots

Autonomous Driving Benchmark: Developed Carla DI-simulator based benchmark for training and testing learned autonomous driving policies (Open-source in GitHub)
- **LLM-powered Decision Agent:** Built AI agent with prompt engineering for human-robot teaming in Overcook-AI environment
- **Bio-inspired Turtle Robot:** Designed mechanical system, motor drive board (ODrive-S1), and control adaptations
- **Django-based Website:** Integrated video conferencing, online chat, and automatic voice transcription

Service and outreach

Paper Review Service 2023 – Present

- IEEE Robotics and Automation Letters (RAL) 2025
- Journal of Field Robotics 2025
- IEEE Transactions on Systems, Man and Cybernetics: Systems 2024, 2025
- International Journal of Human-Computer Interaction 2025
- Journal of Urban Planning and Development 2024
- ACM/IEEE International Conference on Human-Robot Interaction 2024
- IEEE International Conference on Advanced Intelligent Mechatronics 2023

Conference Volunteer
Robotic Science & Systems (RSS) 06/2025

Demos and Field Trips
Demo at Lunar and Planetary Science Conference (LPSC), live robot demo of sensing during locomotion 12/2024
Demo at NASA Ames center, live robot demo of sensing during locomotion and rover travresal 12/2024
Field trip, testing robot sensing, locomotion, and operation at Mt Hood OR, NM 08/2023
Field trip, testing robot sensing, locomotion, and operation at White Sands, NM 03/2023

Press coverage

National & Feature Articles
- **Universe Today**: Walking Moon Robots Possibly More Reliable than Lunar Rovers 04/2025

- **OPB**: How a dog-like robot is training for space exploration on Mount Hood 07/2024
- **KGW 8 - Portland, OR**: Researchers using a four-legged robot on Mount Hood to help them land on the moon 06/2024
- **Koin 6 - Portland, OR**: Robot dog training to walk on the moon on Mt. Hood 04/2024
- **KATU 2 - Portland, OR**: NASA tests walking robot on Mount Hood for space exploration with universities 04/2024
- **CBS News - Los Angeles, CA**: NASA and USC Robotics Team Up to Strengthen Space Exploration Program 04/2024
- **WFMY News 2 - Greensboro, NC**: Robot Dog preparing to head to the moon 04/2024
- **CBS Chicago**: Robotic dog testing out surface for future exploration on the moon 04/2024
- **IEEE Spectrum**: Video Friday: LASSIE On the Moon 04/2024
- **Futurism**: NASA quietly training robot dog to navigate landscape of the moon 04/2024
- **Reuters**: Scientists train a robot to walk on the moon 04/2024
- **NPR Weekend Edition**: A robot dog is training on Earth to be able to go to space one day 04/2024

International Coverage

- **Wired (Spanish)**: Estos investigadores entrenan a robots a caminar sobre terrenos difíciles 04/2024
- **CBC Kids**: A new kind of rover: Dog-like robot trained to explore the moon 05/2024
- **BBC**: Robot dog trains to walk on Moon in Oregon trials 04/2024
- **BBC**: Engineers in Oregon train dog robot to walk on Moon 04/2024
- **CBC**: Meet Spirit, a robot being trained to walk on the moon 04/2024
- **7News Australia**: Robot dog trained to walk on the moon 04/2024
- **Channel 26 Argentina**: Científicos entrenan un robot para caminar sobre la Luna 04/2024
- **El Sol de Mexico**: Robots: From home assistants to planet explorers 04/2024

Institutional Coverage

- **PennToday**: Teaching doglike robots to walk on the moon's dusty, icy surface 04/2024
- **USC Viterbi**: Teaching robots to walk on the moon, and maybe rescue one another 03/2024
- **Texas A&M**: Practice Makes Perfect: Teaching Robots To Walk On The Moon 03/2024
- **Georgia Tech**: Good Dog: LASSIE Spirit Learns to Walk on the Moon 03/2024